

BADSHA LASKAR

Senior Software Engineer | Full Stack AI Developer

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PROFESSIONAL SUMMARY

Senior Software Engineer & Full Stack AI Developer with 4.5+ years of experience architecting and delivering scalable web applications, intelligent AI/ML systems, and data-intensive pipelines for enterprise and international clients. Expert across the entire stack from pixel-perfect React/Next.js frontends and high-performance Node.js/FastAPI backends, to LLM-powered RAG systems, MLOps pipelines, and cloud-native deployments on GCP and Azure. Proven track record of reducing latency, accelerating delivery cycles, and driving measurable business outcomes for UK & US clients.

TECHNICAL SKILLS

Frontend	React, Next.js, Angular, TypeScript, JavaScript (ES6+), Redux Toolkit, Zustand, Tailwind CSS, Shadcn/UI, Vue.js, Angular, React Native, NativeWind, Framer Motion, Webpack, Vite
Backend	Node.js, Express.js, NestJS, Python, FastAPI, Django REST, GraphQL, WebSockets, gRPC, REST APIs, JWT, OAuth2, RBAC, GO
AI / ML	LangChain, LlamaIndex, OpenAI API, HuggingFace Transformers, RAG Pipelines, Vector Databases (Pinecone, Weaviate, pgvector), Prompt Engineering, Fine-tuning (LoRA/QLoRA), Embeddings, Semantic Search
MLOps	MLflow, Weights & Biases, DVC (Data Version Control), Model Registry, A/B Model Testing, Feature Stores, Automated Retraining Pipelines, Model Monitoring & Drift Detection
Databases	PostgreSQL, MongoDB, Redis, Elasticsearch, MySQL, Prisma ORM, SQLAlchemy, Database Schema Design, Query Optimization, Indexing Strategies
DevOps / Cloud	Docker, Kubernetes (basics), GitHub Actions CI/CD, GCP (Cloud Run, GCS, Vertex AI), Azure (Azure ML, App Service), Terraform (basics), Render, Nginx, Linux, Prometheus, Grafana
Other	System Design, Agile/Scrum, TDD/BDD, Code Reviews, Mentoring, Technical Estimation, OpenAPI/Swagger, Postman

EXPERIENCE

SDE II — Senior Full Stack AI Developer

April 2026 – Present

Newton School of Technology

Sonipat, India

- **Architected and shipped a production-grade AI-powered Learning Management System (LMS)** serving 10,000+ students, integrating LLM-driven content generation, adaptive quizzing, and semantic search using **LangGraph**, **pgvector**, and **OpenAI GPT-4o**.
- **Designed and implemented a microservices-based backend** with **NestJS** and **FastAPI**, establishing domain-driven service boundaries, async task queues with **Bull/Redis**, and event-driven communication via **WebSockets** and **Server-Sent Events**.
- **Built a real-time AI code evaluation pipeline** leveraging function calling and structured output schemas, reducing manual grading effort by **70%** and providing instant, contextual feedback to students.
- **Engineered advanced RAG pipelines** with hybrid dense-sparse retrieval (**BM25 + embeddings**), reranking (**RRF**), and contextual compression, improving answer relevance scores by **~40%** over naive retrieval baselines.
- **Spearheaded engineering best practices** - defined coding standards, architectural decision records (ADRs), PR templates, and conducted weekly knowledge-sharing sessions for a team of **8 engineers**.

Software Engineer — Full Stack Developer

March 2022 – April 2026

Zensar Technologies Ltd.

Pune, India

- **Owned end-to-end development of scalable full-stack applications** for **UK & US enterprise clients**, with features used by **thousands of users across production environments** daily.
- **Architected React-based frontend systems** with lazy loading, code splitting, memoization, and virtualized lists, improving perceived UI performance by **~25%** and Time-to-Interactive by **18%**.

- **Designed and deployed containerized Node.js microservices using Docker** with multi-stage builds and environment parity, cutting deployment and setup time by **40%** across dev, staging, and production.
- **Implemented secure authentication & authorization (JWT, RBAC, refresh token rotation)**, reducing login-related support tickets by **50%** and achieving compliance with enterprise security requirements.
- **Built and maintained AI-driven backend workflows** using **LangChain**, **vector databases (Pinecone)**, and **streaming LLM APIs**, accelerating internal feature prototyping velocity by **35%**.
- **Optimized backend data pipelines** with server-sent event streaming, connection pooling, and async processing, cutting AI-related data-processing latency by **45%**.
- **Led CI/CD automation with GitHub Actions** — integrated test gates, coverage reporting, and staged rollouts, shortening release cycles by **30%** and raising test coverage to **~90%**.
- **Designed PostgreSQL schema optimizations** including composite indexes, materialized views, and query plan analysis, reducing p95 API response times by **35%** on data-heavy endpoints.
- **Acted as technical owner** of an internal analytics dashboard (**Angular + Node.js + PostgreSQL**), improving cross-team debugging visibility by **60%** and replacing a manual reporting workflow.
- **Mentored 3 junior engineers**, conducting structured code reviews, pairing sessions, and onboarding, resulting in measurable ramp-up improvement and higher PR quality.

BACKEND, SYSTEM DESIGN & MLOPS HIGHLIGHTS

- Designed **event-driven microservices architectures** with async task queues (Bull/Redis), message brokers, and circuit breakers for high-availability systems.
- Built **RAG pipelines** with multi-stage retrieval: dense vector search (pgvector / Pinecone), BM25 sparse search, reranking (Cohere Rerank), and contextual compression for production LLM apps.
- Implemented **MLOps workflows** using MLflow for experiment tracking, model versioning, and registry management; integrated drift monitoring to trigger automated retraining on performance degradation.
- Engineered **Feature Store patterns** for ML feature reuse across training and serving environments, reducing data inconsistency between offline training and online inference.
- Designed **scalable REST & GraphQL API architectures** with OpenAPI documentation, rate limiting, request validation (Zod/Pydantic), and observability via structured logging and distributed tracing.
- Implemented **database performance strategies** partitioning, indexing, caching layers (Redis), and query optimization for high-throughput, low-latency data access patterns.

CLOUD & INFRASTRUCTURE EXPOSURE

- Deployed **AI inference endpoints on GCP Cloud Run and Vertex AI**, leveraging auto-scaling and GPU-backed instances for cost-efficient LLM serving.
- Configured **Azure ML workspaces** for experiment tracking and model deployment, integrating with Azure App Service for seamless full-stack deployments.
- Managed **multi-environment Docker Compose configurations** with secrets management via environment injection and GitHub Actions secrets vaulting.
- Set up **Prometheus + Grafana dashboards** for real-time API health monitoring, latency percentile tracking, and alerting on error rate spikes.

PROJECTS

AI Note & Board Question Generator [Live Demo](#)

- Built a **production-grade AI SaaS** with RAG-based retrieval (pgvector + OpenAI embeddings), structured output generation (GPT-4o function calling), and multi-format export — handling **2,500+ documents** in testing.
- Architected a **FastAPI backend with async SQLAlchemy**, background task queues, streaming responses, and a React (TanStack Query) frontend with optimistic UI and real-time progress updates.
- Integrated **MLflow for LLM call tracing** and prompt versioning, enabling reproducible experimentation and quality benchmarking across model versions.
- Implemented **CI/CD with GitHub Actions**, Dockerized multi-service deployment, and automated integration tests — simulating real-world SaaS delivery workflows.

React Native Mobile Writing App — QuilKalam [Live Demo](#)

- Designed and shipped a **distraction-free writing app** with React Native + NativeWind achieving a **great user rating**, with rich-text editing, offline sync, and cloud backup.
- Implemented **offline-first architecture** with SQLite local storage and **Next.js SSR backend** for cloud sync, supporting 100+ active users with reliable cross-device state management.

Shortest Path Algorithm Visualizer [Live Demo](#)

- Built an interactive algorithm visualizer using **Next.js 14 and DaisyUI**, demonstrating DFS, BFS, Dijkstra, and A* with real-time step-by-step animation and configurable grid parameters.

Sorting Algorithm Visualizer [Live Demo](#)

- Built an interactive, visually appealing user interface using **JavaScript and React**
- Implemented and showcased key sorting algorithm such as **Quick sort, Merge sort, Bubble sort etc.**

EDUCATION

Bachelor of Technology — Computer Science & Engineering

Techno India University, Kolkata, India